

Palm Fruit Ripeness Detection and Counting Using YOLOv8 Algorithm in PTPN IV Medan North Sumatera Indonesia

Rael Russel Hutapea
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
rael.hutapea@binus.ac.id

Samdo Jumar Purba
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
samdo.purba@binus.ac.id

William Ferdinandz Tandion
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
william.tandion@binus.ac.id

Erna Fransisca Angela Sihotang
Statistics Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
erna.sihotang@binus.ac.id

Edy Irwansyah
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
edirwan@binus.ac.id

Abstract—Oil palm plantations have become a crucial sector in the Indonesian economy. This research investigates the effectiveness of YOLOv8 in identifying the ripeness of oil palm fruits. The goal is to create a system that uses the YOLOv8 Deep Learning algorithm to detect the ripeness level of oil palm fruits. An automatic detection system will simplify and accelerate the sorting or classifying of oil palm fruits. This research utilizes datasets sourced from the oil palm plantation repository of PTPN IV Medan, North Sumatera, Indonesia. The dataset collected for this research consists of 85 images with a size of 640x640. The dataset is loaded into Roboflow and undergoes pre-processing and augmentation. The total dataset obtained after augmentation is 185 images. After obtaining the API from Roboflow and integrating it into Google Colab for YOLOv8 training, testing was performed on the provided images. The results indicated that YOLOv8 achieved an accuracy of 0.991 (99.1%) in detecting ripe, unripe, and rotten oil palm fruit. The high value of accuracy leads to the potential to increase the efficiency of oil palm plantation management and become the basis for developing AI-based agricultural technology. With further updates, the YOLOv8 model could become a handy tool in oil palm plantation monitoring and management in the future.

Keywords—YOLOv8, palm fruit, object detection, deep learning, PTPN IV

I. INTRODUCTION

Indonesia has rich natural resources. Indonesia is a major global producer country of palm oil. Palm fruit, once processed into vegetable oil, can be used in a variety of everyday products such as ice cream, chocolate, biscuits, cosmetics, soap, detergents, textile oils, and biodiesel. [1]. Oil palm plantations have become a crucial sector in the Indonesian economy. By 2020, the area of oil palm plantations in North Sumatera is expected to cover 440,000 hectares, with a production capacity of 7 million tons. [2]. One of the main entities that contributes significantly to the oil palm industry is PT Perkebunan Nusantara (PTPN) IV, located in Medan, North Sumatera, Indonesia. PTPN IV is a state-owned enterprise that focuses on the oil palm business which has now been organizationally united and become subholding of PalmCo. The oil palm commodity is so important that efforts are needed to improve it, both in terms of

productivity and efficiency. To enhance this, a key factor is the accurate measurement of oil palm fruit ripeness. Rapid and precise assessment of fruit ripeness can lead to higher harvest yields.

Computer vision is a branch of artificial intelligence dedicated to creating systems that can analyze, understand, and interpret visual data from the real world. One model of this technology is You Only Look Once (YOLO), the model chosen in this research is YOLO version 8. Thanks to its fast-processing method, the YOLO model offers benefits such as high-accuracy detection and real-time performance. Previously, YOLO models have been used to identify agricultural fruits like apples, tomatoes, and pears. In the oil palm sector research, earlier studies created an automated detection system incorporating the YOLO model to identify FFBs. The authors evaluated the performance of different YOLOv3 series models, and the findings demonstrated that the model is effective for object detection in the oil palm sector [3]. This research uses the YOLOv8 model because YOLOv8 is the newest version of You Only Look Once, offering enhanced speed and accuracy in detection. It utilizes a similar backbone to YOLOv5 but includes modifications to the CSPLayer, now referred to as the C2f module. The C2f module (cross-stage partial bottleneck with two convolutions) integrates high-level features with contextual information to enhance detection accuracy [4].

The goal of this research is to create a system for assessing the ripeness of oil palm fruits using the YOLOv8 Deep Learning algorithm. When detecting oil palm fruits manually, the main challenge is undoubtedly the need for more time. An automatic detection system will simplify and accelerate the sorting or classifying of oil palm fruits. Improvements in terms of productivity must continue to be carried out to obtain maximum results. With increased productivity, harvest yields can certainly increase significantly. The YOLOv8 technology is expected to contribute significantly to speeding up the detection of oil palm fruits.

The writing structure of this paper consists of several sections. Firstly, the introduction provides information about the superiority of the algorithm used and the purpose of the

research. Then, the state-of-the-art section contains the latest research in the detection of oil palm fruit ripeness using the YOLO model. Following that, the methodology section elaborates on the research flowchart, data, and YOLOv8 algorithm model. Then, the results and discussions section present the findings and analysis. Finally, the paper concludes with the conclusions section.

II. STATE OF THE ART

A considerable amount of research has been conducted on detecting the ripeness of oil palm fruit using various models, including YOLO. This includes several previous studies that utilized the YOLO model to assess the ripeness of oil palm fruits. This paper [2] compares three object detection models, namely MobileNetV2 SSD, EfficientDet, and YOLOv5, in categorizing the ripeness stages of fresh oil palm fruit. The results show that YOLOv5m has the highest mean average precision (mAP) value of 84.2%. YOLOv5's advantages include faster detection speed, high computational efficiency, and the ability to be implemented on mobile devices. However, although EfficientDet is a powerful object detector, it has not demonstrated similar performance to YOLOv5. Meanwhile, MobileNetV2 SSD, although intended for mobile applications, has not performed as well as other models in object detection. Thus, YOLOv5m is proposed as the most suitable object detection model for oil palm fresh fruit ripeness classification applications, with opportunities for future improvement through hyperparameter adjustments and the implementation of ensemble model techniques. This research highlights the significance of digitalizing the agricultural industry and applying artificial intelligence (AI) in agricultural practices, intending to improve efficiency and productivity in oil palm plantation management by an appropriate object detection model.

The [3] brings innovation in the development of a real-time ripe oil palm fruit detection system based on the YOLOv4 model. By utilizing a training dataset that includes 430 positive images and 250 negative images and setting the model parameters to a width and height of 608, this research managed to achieve an impressive level of accuracy. Field testing showed excellent performance, with mAP results reaching 87.9%, confirming the system's ability to identify ripe palm fruit in real-time. This makes an important contribution in the context of modern agriculture and the plantation industry, providing a strong foundation for the development of automated detection technology that can help increase efficiency and productivity in the process of packing ripe palm fruit.

In the research conducted by [5], the YOLOv3 algorithm is used to detect ripe palm fruit in real-time, to improve the harvest process. The method involves data collection, image labelling, model training, and testing using the Darknet framework. The research results show that YOLOv3 can detect and differentiate the ripeness levels of oil palm fruits with relatively high accuracy, with a Mean Average Precision (mAP) value reaching 95.18% at the 5000th iteration. However, there is a decrease in accuracy after the 5000th iteration due to overfitting. There are also some drawbacks, including the lack of high-quality images in the dataset, which affects the detection accuracy. This research shows the potential of the YOLOv3 algorithm in increasing oil palm

harvest efficiency, however further improvements in dataset collection and quality are needed to increase overall detection accuracy.

In [6], focused on detecting and classifying the ripeness of oil palm fruit using different variations of the YOLOv8 model. Utilizing a comprehensive dataset, the study demonstrates that the YOLOv8m model achieves the highest accuracy, with a mAP value of 92.7%. Further testing also showed that YOLOv8 able to detect objects with a high accuracy of 99.49%. This reflects an excellent potential of the model in changing the paradigm of evaluating oil palm fruit ripeness, which traditionally relies on human graders, into a more consistent and accurate process, as well as minimizing economic losses due to suboptimal oil extraction rates caused by unripe fruits. Thus, the YOLOv8 model can be an effective tool in the oil palm industry, which can have a positive impact on the productivity and sustainability of the industry. This paper also highlights the potential of the YOLOv8 model in general agricultural applications, by showing its possible use in different agricultural contexts.

Here are the previous research studies with their respective algorithms used for detecting the ripeness of oil palm fruits, as shown in Table I.

TABLE I. PREVIOUS RESEARCH ON OIL PALM RIPENESS CLASSIFICATION USING MACHINE LEARNING AND DEEP LEARNING

No	Model/Method/Algorithm	Dataset	Accuracy
[1]	Otsu Method - Support Vector Machines (SVM)	160 images (Raw, Under-ripe, Ripe)	92.5%
[2]	YOLOv5	304 images (Unripe, Halfripe, Ripe, Over-ripe)	84.2%
[3]	YOLOv4	490 images (half containing Ripe, and half not containing Ripe)	87.9%
[5]	YOLOv3	4500 images, (Unripe, Ripe, Overripe)	95.18%
[6]	YOLOv8	14559 objects within 7171 images (Unripe, Underripe, Ripe, Overripe, Empty Bunch, Abnormal)	99.49%
[7]	Convolutional Neural Network (CNN)	400 images (Raw, Ripe, Rotten)	92%
[8]	contour-based approach	160 images (Raw, Under-ripe, Ripe)	90.13%
[9]	YOLOv4	14559 objects within 7171 images (Unripe, Underripe, Ripe, Overripe, Empty Bunch, Abnormal)	99%
[10]	Gray Level Co-occurrence Matrix (GLCM)	99 images (Raw, Ripe, Overripe)	71.4%
[11]	Color Features Extraction and Bag of Visual Words method	400 images (Ripe, Unripe, Underripe, Overripe)	57% and 70%
[12]	YOLOv8 and EfficientDet Lite-4	8299 images (Unripe, Underripe, Ripe, Overripe, Empty Bunch, Abnormal)	97.6% and 94%
[13]	ResNet50 and YOLOv3	299 images (Underripe, Ripe, Overripe)	76% and 84%
[16]	Convolutional Neural Network (CNN) - EfficientNetB0	328 images (Unripe, Underripe, Ripe)	91.4% to 91.5%

No	Model/Method/Algorithm	Dataset	Accuracy
[17]	Linear Discriminant Analysis (LDA)	150 images (Raw, Under-ripe, Ripe)	98.88%
[18]	Artificial Neural Network (ANN) - Back-Propagation Algorithm	240 images (Raw, Half-ripe, Ripe)	98.30%
[19]	Convolutional Neural Network (CNN) - AlexNet and DenseNet	400 images (Ripening, Raw, Less Ripped, Almost Ripped, Ripped, Perfectly Ripped, Too Ripped)	77% and 86%
[20]	Convolutional Neural Network (CNN)	628 images (Unripe, Ripe)	98%

III. METHODOLOGY

In this research, there are several steps that can be seen in Figure 1. The initial process of this research involves observing the problem to be identified, followed by conducting a literature review related to the problem to be solved. Next is data collecting, which is carried out at PTPN IV Medan, North Sumatera, Indonesia. After the data is collected, pre-processing and post-processing are performed to train the model. Then, the model produces results that are subsequently evaluated to obtain the desired outcome, which is then analyzed to reach a conclusion.

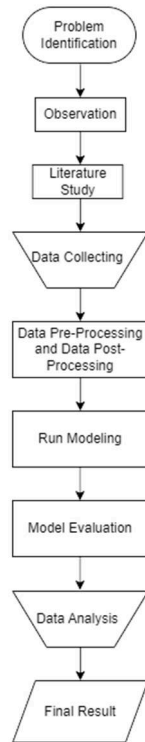


Fig. 1. Research Flowchart

A. Data Collecting

A dataset from the PTPN IV oil palm plantation repository is used in this research. To maximize the data outcomes, the data collection activities were conducted in an organized manner. Various techniques were employed to gather data, including conducting interviews with oil palm employees, farmers, and office workers. Finding out about the present field conditions, when to harvest oil palm fruit, issues encountered in the field, cultivation techniques, and technologies employed are all part of this interview. Whether

the oil palm fruit is ripe, unripe, or rotten, activities are conducted in an organized fashion.

The speed and accuracy of oil palm fruit detection will increase with a clean dataset and clear image quality. An iPhone 12 was used as the mobile device camera to capture and organize images of oil palm fruit for this dataset.



Fig. 2. Dataset Example from the PTPN IV oil palm plantation

It can be seen in Figure 2 that various types of images were collected, that includes images of ripe oil palm fruit, unripe oil palm fruit, rotten oil palm fruit, and images of several oil palm fruits. To achieve accurate detection and to enable detection, images must be taken in a clear and organized manner. A dataset consisting of 85 photos with an image size of 640x640 was collected for this research. The YOLOv8 model will then be trained and used to detect the obtained dataset.

B. Image Pre-Processing

The three steps of the data pre-processing procedure are uploading, annotating, and foldering. The collected image dataset was uploaded to Roboflow, then divided into good and bad images in the foldering process, as well as photos that clearly show fruit, which will be used as training data later. Then, the image will be separated into three sections: ripe, unripe, and rotten. Following their division into these groups, the images will receive annotations or labels according to these categories. After that, each image that has been annotated and labeled will be divided into 3 sets: a training set with 63 images (88% ratio), a validation set with 13 images (7% ratio), and a testing set with 9 images (5% ratio). These images are divided randomly by Roboflow but can be edited or modified by the user. Roboflow offers various pre-processing characteristics that are applied to the dataset. These features boost the efficiency of data training and decrease training time by shrinking the existing data.

C. Image Post-Processing

Post-processing focuses on enhancing the segmented image, and any additional processing needed is handled during this phase [14]. In this step, a few changes are made to the dataset to ensure that the database has more data variations. This can be achieved with the augmentation feature of Roboflow, where users can select the type of augmentation, and the image will be transformed based on that augmentation. Two types of augmentation are used in the available dataset: Flip and 90° Rotate. The total dataset obtained after augmentation is 185 images, with the number of images in the training set increasing to 163 images.

D. YOLOv8 Model

Developed by Ultralytics, the same company behind YOLOv5, YOLOv8 is the latest advanced computer vision model. It offers built-in capabilities for object recognition, classification, and segmentation tasks. YOLOv8 can be accessed via the command line interface and the Python package [6]. YOLOv8 is a great algorithm for various object recognition and tracking, instance segmentation, image classification, and pose estimation tasks due to its fast, precise, and user-friendly architecture. In this project, YOLOv8 will be used as the main algorithm for several reasons such as:

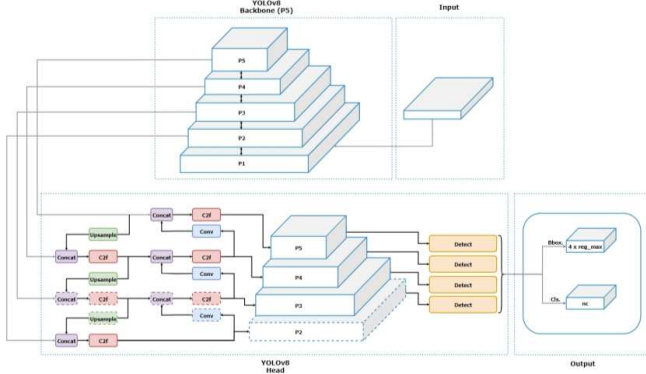


Fig. 3. YOLOv8 Architecture [15]

1) *Enhanced CSPDarknet Backbone*: YOLOv8 utilizes an improved version of the CSPDarknet backbone, which stands for Cross Stage Partial Darknet. This backbone is responsible for extracting hierarchical features from input images. Key features include:

a) *CSP (Cross Stage Partial) Connections*: These connections help in reducing computational complexity while maintaining gradient flow, allowing the network to learn more effectively.

b) *Bottleneck Layers*: YOLOv8 incorporates advanced bottleneck layers that include convolutional operations, activation functions (like Mish or SiLU), and normalization layers (such as BatchNorm or GroupNorm).

c) *Residual Connections*: Facilitates the training of deeper networks by allowing gradients to flow through shortcut connections, mitigating the vanishing gradient problem.

2) *Feature Pyramid Integration*: The backbone processes the input image through multiple stages, each capturing different levels of abstraction. Lower layers capture fine-grained details, while higher layers capture more abstract, semantic information.

IV. RESULTS AND DISCUSSION

YOLOv8 model training was carried out in Google Colab using the dataset that had been uploaded to Roboflow and had undergone pre-processing and augmentation. After obtaining the dataset API from Roboflow and integrating it into Google Colab for YOLOv8 training, testing was conducted on the provided images.

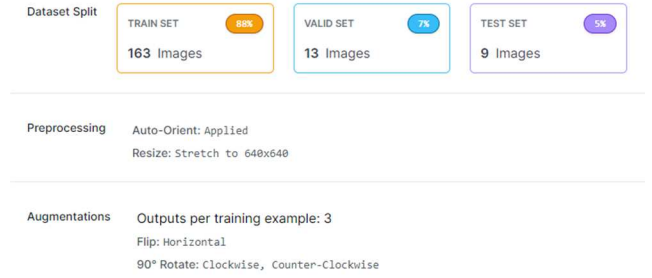


Fig. 4. Details of pre-processing and augmentation

Figure 4 shows the details of pre-processing and augmentation of the oil palm fruit dataset. Pre-processing involves using Auto-Orient to adjust the visual orientation of the images to match how they are stored on disk and resizing images to 640x640. Data augmentation uses 2 variations: Flip and 90° Rotate.

A. Training YOLOv8 Model

YOLOv8 model training was carried out for 25 epochs with an image size of 640x640. Below is a training graph of YOLOv8 that shows changes during training process over each epoch shown in Figure 5.

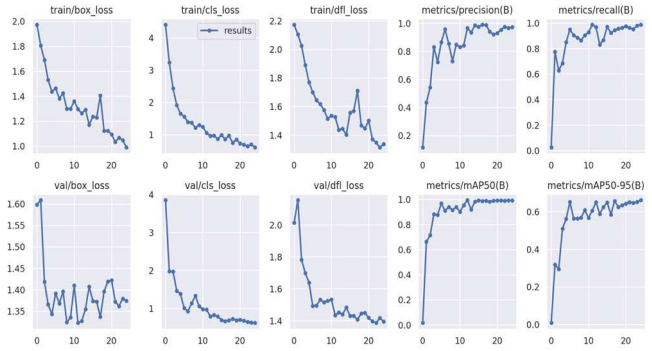


Fig. 5. YOLOv8 Training Graph

TABLE II. YOLOv8 TRAINING RESULTS

Class	mAP50	mAP50-95
All	0.991	0.662
Ripe	0.988	0.606
Unripe	0.995	0.717

Table II shows the accuracy of each class after the training process has been carried out. In this research, it is divided into 3 classes: Ripe, Unripe, and Rotten. It was found that YOLOv8 had an accuracy of 0.991 (99.1%) in detecting all the three classes (ripe, unripe, and rotten) of oil palm fruit. Ripe has mAP50 (mean average precision) of 0.988 and 0.606 with the higher threshold. Unripe has mAP50 of 0.995 and 0.717 with higher thresholds. Cross validation was used in these tests, and the results are almost the same but not identical.

B. Detection Results

In the detection process, oil palm fruits were successfully identified with three main categories: ripe oil palm fruit, unripe oil palm fruit, and rotten oil palm fruit. As shown in Figure 6 is an example of research results on the detection of ripe oil palm fruit. As shown in the figure, the detection results

are correct because the trained model detects the ripe oil palm fruit precisely and accurately with arguably high confidence. While Figure 7 shows unripe oil palm fruit detection results. It represents an example of research results on the detection of unripe oil palm fruit. As shown in the figure, the detection results are correct because the trained model detects the unripe oil palm fruit precisely and accurately.



Fig. 6. Ripe oil palm fruit detection results



Fig. 7. Unripe oil palm fruit detection results

Then, on Figure 8 is an example of research results on the detection of rotten oil palm fruit. As shown in the figure, the detection results are correct because the trained model detects the rotten oil palm fruit precisely and accurately. It can be seen that the pink bounding box was correctly placed.

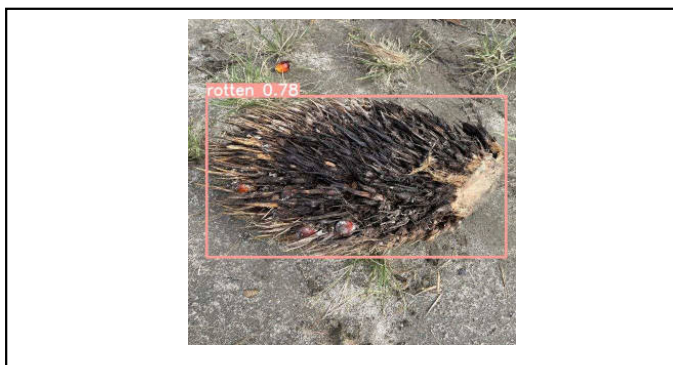


Fig. 8. Rotten oil palm fruit detection results

C. Discussion

According to the findings of this research, the YOLOv8 model, which is the newest version of YOLO, provides better accuracy compared to the previous versions of YOLO used by previous researchers. However, when compared to previous research that used the YOLOv8 model, the accuracy results of this research can be considered quite good, although it can still be improved when compared with similar research that shows very accurate results from using YOLOv8 model

[6] [12]. Even though this research uses a smaller dataset and epochs, the accuracy achieved is not significantly lower compared to studies that used larger dataset and epochs. However, based on the results in Table II, the accuracy results for the rotten class are not shown. This may occur due to insufficient datasets to train the model in the rotten ripeness class. Therefore, it is important to collect sufficient and large datasets for each ripeness class, so the model can be more effectively trained in detecting the ripeness level of oil palm fruit. To achieve high accuracy results, one effective way is by providing/collecting sufficient and large datasets that includes clear and well-defined images for each ripeness level. Furthermore, training the model with more epochs allows it to improve its understanding of the dataset, thereby improving its ability to generalize and make accurate predictions. By combining large datasets with extended training, the model can achieve better performance in detecting the ripeness level.

V. CONCLUSION

Based on this research, it can be concluded that the YOLOv8 model has been effective in detecting the ripeness of oil palm fruits at PTPN IV Medan, North Sumatera, Indonesia. This model successfully and accurately identifies the ripeness level of the oil palm fruits with an accuracy of 0.991 (99.1%). The results obtained show the potential to increase the efficiency of oil palm plantation management and can serve as a basis for further development in AI-based agricultural technology. With additional updates and advancements, the YOLOv8 model could become an extremely valuable tool for monitoring and managing oil palm plantations in the future.

ACKNOWLEDGEMENT

We thank ESRI Indonesia and PTPN VI as the basis for obtaining data and programming tools to obtain the results of this research.

REFERENCES

- [1] A. Septiarini, H. Hamdani, H. R. Hatta and A. A. Kasim, "Image-based processing for ripeness classification of oil palm fruit," *2019 5th International Conference on Science in Information Technology (ICSITech)*, pp. 23-26, 2019.
- [2] M. Y. M. A. Mansour, K. D. Dambul, and K. Y. Choo, "Object Detection Algorithms for Ripeness Classification of Oil Palm Fresh Fruit Bunch," *International Journal of Technology*, vol. 13, no. 6, pp. 1326-1335, 2022.
- [3] J. W. Lai, H. R. Ramli, L. I. Ismail, and W. Z. W. Hasan, "Real-Time Detection of Ripe Oil Palm Fresh Fruit Bunch Based on YOLOv4," *IEEE Access*, vol. 10, pp. 95763-95770, 2022.
- [4] J. Terven, D. M. Córdova-Esparza, and J. A. Romero-González, "A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *Machine Learning and Knowledge Extraction*, vol. 5, no. 4, pp. 1680-1716, 2023.
- [5] N. A. Mohd Basir Selvam, Z. Ahmad and I. A. Mohtar, "Real Time Ripe Palm Oil Bunch Detection using YOLO V3 Algorithm," *2021 IEEE 19th Student Conference on Research and Development (SCORED)*, pp. 323-328, 2021.
- [6] T. S. Gunawan, M. Kartiwi, H. Mansor, and N. M. Yusoff, "Palm Fruit Ripeness Detection and Classification Using Various YOLOv8 Models," *2023 IEEE 9th International Conference on Smart Instrumentation, Measurement and Applications (ICSIMA)*, pp. 193-198, 2023.
- [7] S. Ashari, G. J. Yanris, and I. Purnama, "Oil Palm Fruit Ripeness Detection using Deep Learning," *Sinkron*, vol. 7, no. 2, pp. 649-656, 2022.

- [8] A. Septiarini, H. Hamdani, H. R. Hatta, and K. Anwar, "Automatic image segmentation of oil palm fruits by applying the contour-based approach," *Scientia Horticulturae*, vol. 261, p. 108939, 2020.
- [9] Suharjito, F. A. Junior, Y. P. Koeswandy, Debi, P. W. Nurhayati, M. Asrol and Marimin, "Annotated Datasets of Oil Palm Fruit Bunch Piles for Ripeness Grading Using Deep Learning," *Scientific Data*, vol. 10, no. 1, p. 72, 2023.
- [10] A. Syaifuddin, L. N. A. Mualifah, L. Hidayat, and A. M. Abadi, "Detection of palm fruit maturity level in the grading process through image recognition and fuzzy inference system to improve quality and productivity of crude palm oil (CPO)," *Journal of Physics: Conference Series*, vol. 1581, no. 1, p. 012003, 2020.
- [11] S. A. Ghazali, H. Selamat, Z. Omar, and R. Yusof, "Image Analysis Techniques for Ripeness Detection of Palm Oil Fresh Fruit Bunches," *ELEKTRIKA- Journal of Electrical Engineering*, vol. 18, no. 3, pp. 57-62, 2019.
- [12] M. R. A. Priyadi and Suharjito, "Comparison of YOLOv8 and EfficientDet4 Algorithms in Detecting the Ripeness of Oil Palm Fruit Bunch," *2023 10th International Conference on ICT for Smart Society (ICISS)*, pp. 1-7, 2023.
- [13] N. Khamis, H. Selamat, S. Ghazalli, N. I. M. Saleh, and N. Yusoff, "Comparison of Palm Oil Fresh Fruit Bunches (FFB) Ripeness Classification Technique using Deep Learning Method," in *2022 13th Asian Control Conference (ASCC)*, pp. 64-68, 2022.
- [14] P. Shivhare and V. Gupta, "Review of Image Segmentation Techniques Including Pre & Post Processing Operations," *International Journal of Engineering and Advanced Technology*, vol. 4, no. 3, pp. 153-157, 2015.
- [15] N. B. A. Karna, M. A. P. Putra, S. M. Rachmawati, M. Abisado and G. A. Sampedro, "Toward Accurate Fused Deposition Modeling 3D Printer Fault Detection Using Improved YOLOv8 With Hyperparameter Optimization," *IEEE Access*, vol. 11, pp. 74251-74262, 2023.
- [16] W. Y. Teh and I. K. T. Tan, "Coloured Edge Maps for Oil Palm Ripeness Classification," *British Machine Vision Conference*, 2021.
- [17] A. Septiarini, H. R. Hatta, H. Hamdani, A. Oktavia, A. A. Kasim and S. Suyanto, "Maturity Grading of Oil Palm Fresh Fruit Bunches Based on a Machine Learning Approach," *2020 Fifth International Conference on Informatics and Computing (ICIC)*, pp. 1-4, 2020.
- [18] A. Septiarini, A. Sunyoto, H. Hamdani, A. A. Kasim, F. Utaminigrum, and H. R. Hatta, "Machine vision for the maturity classification of oil palm fresh fruit bunches based on color and texture features," *Scientia Horticulturae*, vol. 286, p. 110245, 2021.
- [19] H. Herman, T. W. Cenggoro, A. Susanto, and B. Pardamean, "Deep Learning for Oil Palm Fruit Ripeness Classification with DenseNet," *2021 International Conference on Information Management and Technology (ICIMTech)*, vol. 1, pp. 116-119, 2021.
- [20] A. Y. Saleh and E. Liansitim, "Palm oil classification using deep learning," *Science in Information Technology Letters*, vol. 1, no. 1, pp. 1-8, 2020.